



Satyam Kumar

Roll No.: 23PHC1R45

Master of Science and Technology

Engineering Physics

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Portfolio Profile

GitHub Profile

LinkedIn Profile

EDUCATION

• National Institute of Technology, Warangal

MS in Engineering Physics

CGPA: 7.05

2023 – Present

• Vinoba Bhave University

BS in Physics and Mathematics

CGPA: 7.35

2019 – 2022

PROJECTS

• DocQA – Chat with PDFs | FastAPI, LangChain, LlamaIndex, FAISS

[Live](#) / [Source Code](#)

- Built an end-to-end **RAG (Retrieval-Augmented Generation)** system enabling users to query PDF documents using LLM-based contextual retrieval.
- Implemented document ingestion pipeline: PDF chunking, embeddings generation, and indexing using **FAISS vector store**.
- Utilized **LangChain & LlamaIndex** to orchestrate prompt templates, retrieval chains, and query routing for improved answer accuracy.
- Designed a FastAPI backend with optimized I/O, async processing, and PostgreSQL metadata storage.

• PixLune – AI Image Generation Platform | Stable Diffusion, MERN Stack

[Live](#) / [Source Code](#)

- Built an AI image generation system using **Stable Diffusion XL Base 1.0** via HuggingFace Inference API.
- Created inference workflow supporting prompts, styles, negative prompts, and multi-resolution image generation.
- Designed secure backend pipelines for user requests, including JWT authentication and rate-limited model generation.
- Integrated **Cloudinary** for storage and optimized CDN delivery of generated assets.

• AlgoVisualizer – Sorting Visualizer | React, Vite, Tailwind

[Live](#) / [Source Code](#)

- Built an interactive visualization tool demonstrating sorting algorithms; used to improve algorithmic intuition.
- Implemented optimized rendering and asynchronous animations for smooth visual performance.

• YouTube Video Downloader Backend | Node.js, Express, ytdl-core

[Source Code](#)

- Developed a backend pipeline for high-speed video downloads using underlying YouTube extraction APIs.
- Implemented request validation, streaming optimization, and error-handled download workflows.

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, TypeScript.

AI/ML Tools: LangChain, LlamaIndex, HuggingFace Transformers, FAISS, ChromaDB, RAG Pipelines, OpenAI API, Vector Databases.

Backend for AI: FastAPI, Node.js, Express.js, REST APIs.

ML/Deep Learning: PyTorch (basic), NumPy, Pandas, Scikit-learn.

Cloud & Deployment: AWS S3, EC2, Docker, Render, Railway, Vercel.

Developer Tools: Git, GitHub, VS Code, Jupyter Notebook.

Core CS Concepts: Data Structures & Algorithms, OOP, OS, CN, DBMS.

CERTIFICATIONS

- **Full Stack Web Development** Full Stack Web Development course by Angela Yu (Udemy). [🔗](#)
- **Full Stack Development Cohort 2.0** Cohort 2.0 Full Stack Development program.
- **CS50 Introduction to Computer Science** Harvard CS50x (OpenCourseWare).
- **Mastering Data Structures and Algorithms** Course by Abdul Bari (Udemy). [🔗](#)
- **Python for Absolute Beginners** Udemy course. [🔗](#)

ACHIEVEMENTS & AWARDS

- **Solved 800+ DSA Problems** Across LeetCode and Codeforces
- **Competitive Ratings** LeetCode: 1630 | Codeforces: 1070 (Newbie)
- **Top 7% in LeetCode Weekly Contest 476** Among 30,000+ participants
- **Top 10% in LeetCode Weekly Contest 464** Among 35,700+ participants
- **Top 15% in LeetCode Bi-Weekly Contest 167** Among 31,773+ participants